

# Hefang Deng

[turbodeng.github.io](https://turbodeng.github.io) | [denghefang@sjtu.edu.cn](mailto:denghefang@sjtu.edu.cn) | [Google Scholar](#) | [github.com/turbodeng](https://github.com/turbodeng)

## RESEARCH INTERESTS

My research interest lies in turbomachinery aerodynamics, next-generation advanced propulsion, and AI for Science. Currently, my research focuses on leveraging AI to advance and elevate aerodynamic design capabilities for advanced turbomachinery.

## EDUCATION

### Shanghai Jiao Tong University

*Doctor of Philosophy in Aerospace Engineering*

- **Supervisor:** Prof. Jinfang Teng

Shanghai, China

*Apr 2021 – Dec 2024*

### Harbin Engineering University

*Master of Engineering in Power Engineering and Engineering Thermophysics*

- **Supervisor:** Prof. Yuting Jiang

Harbin, China

*Sep 2018 – Apr 2021*

### Harbin Engineering University

*Bachelor of Engineering in Energy and Power Engineering*

- **Supervisor:** Prof. Qun Zheng

Harbin, China

*Sep 2014 – Jul 2018*

## RESEARCH PROJECTS

### Shanghai Municipal Education Commission Joint Innovation Program

Research on Tip Leakage Flow Mechanism in High-Pressure Compressors

*Apr 2021 – Dec 2024*

### National Natural Science Foundation of China (General Program)

Evolution and Control Mechanism of Clearance Flow Field in Cantilevered Stator with High-Speed Rotating Hub

*Apr 2021 – Dec 2024*

### National Science and Technology Major Project

Theoretical and Methodological Research on Aerodynamic Stability Analysis of Multi-Stage Compressors

*Apr 2021 – Dec 2024*

### National Natural Science Foundation of China (Major Program)

Research on Fault Precursor Enhancement and Fusion Early Warning Method for Large Civil Aeroengine Ground Test

*Jan 2024 – Dec 2024*

## SELECTED PUBLICATIONS

- [1] **Hefang Deng**, Songan Zhang, Honglin He, Mingmin Zhu, Xiaoqing Qiang, Jinfang Teng. Design optimization methods for a compressor rotor under inlet distortion. *Aerospace Science and Technology*, Vol. 167, 109987, 2026.
- [2] **Hefang Deng**, Huihui Li, Songan Zhang, Mingmin Zhu, Xiaoqing Qiang, Jinfang Teng. Aerodynamic performance in a boundary layer ingesting fan with swirl distortion. *Aerospace Science and Technology*, Vol. 148, 109102, 2024.
- [3] Honglin He, **Hefang Deng**, Xiang Zuo, Songan Zhang, Jinfang Teng, Mingmin Zhu. Comparative study of neural structure paradigms for compressor cascade flow field modeling based on the condition-encoder. *Aerospace Science and Technology*, Vol. 166, 109898, 2026.
- [4] Kailong Xia, **Hefang Deng**, Mingmin Zhu, Huayin Chen, Jinfang Teng. A novel dynamic mesh method for multi-scenario turbomachinery simulations: Development, validation, and application. *Aerospace Science and Technology*, Vol. 169, 110120, 2026.
- [5] **Hefang Deng**, Songan Zhang, Kailong Xia, Xiaoqing Qiang, Mingmin Zhu, Jinfang Teng. Partial rotating instability in a boundary layer ingesting fan. *Chinese Journal of Aeronautics*, Vol. 38, No. 3, 100217, 2025.
- [6] **Hefang Deng**, Songan Zhang, Kailong Xia, Xiaoqing Qiang, Mingmin Zhu, Jinfang Teng. On the mechanism of rotating stall in a boundary layer ingesting fan. *Journal of Engineering for Gas Turbines and Power*, Vol. 147, No. 9, 091018, 2025.

- [7] Yaqi Liu, **Hefang Deng**, Jinfang Teng, Mingmin Zhu, Songan Zhang, Shaopeng Lu. Loss mechanisms of leakage flow in a cantilevered stator with/without moving end wall. *Physics of Fluids*, Vol. 37, No. 3, 035131, 2025.
- [8] **Hefang Deng**, Songan Zhang, Kailong Xia, Xiaoqing Qiang, Mingmin Zhu, Jinfang Teng. Performance prediction and design optimization of a transonic rotor based on deep transfer learning. *Physics of Fluids*, Vol. 36, No. 9, 097107, 2024.
- [9] Kailong Xia, **Hefang Deng**, Yongjian Zhong, Mingmin Zhu, Xiaoqing Qiang, Jinfang Teng. The impact of second-harmonic tip clearance distributions caused by casing ovalization and blade height variation in a large-scale compressor stage. *Aerospace Science and Technology*, Vol. 149, 109180, 2024.
- [10] Xiaoqing Qiang, **Hefang Deng**, Kailong Xia, Jinfang Teng, Mingmin Zhu, Shaopeng Lu. Effects of moving endwall on the unsteadiness of tip leakage flow in compressor cascades. *Physics of Fluids*, Vol. 35, No. 10, 105121, 2023.
- [11] **Hefang Deng**, Kailong Xia, Xiaoqing Qiang, Mingmin Zhu, Jinfang Teng, Huihui Li. Investigation on the inflow distortion of a propulsive fuselage aircraft and its impact on the propulsion fan performance. *Physics of Fluids*, Vol. 35, No. 8, 085122, 2023.
- [12] **Hefang Deng**, Kailong Xia, Jinfang Teng, Xiaoqing Qiang, Mingmin Zhu, Shaopeng Lu. An experimental facility with the high-speed moving endwall for axial compressor leakage flow research. *Aerospace*, Vol. 10, No. 5, 432, 2023.
- [13] **Hefang Deng**, Kailong Xia, Jinfang Teng, Shaopeng Lu, Mingmin Zhu, Xiaoqing Qiang. Loss analysis of cavity leakage flow in a compressor cascade. *Journal of Turbomachinery*, Vol. 144, No. 12, 121008, 2022.
- [14] **Hefang Deng**, Jinfang Teng, Mingmin Zhu, Xiaoqing Qiang, Shaopeng Lu, Yuting Jiang. Overall cooling performance evaluation for film cooling with different winglet pairs vortex generators. *Applied Thermal Engineering*, Vol. 212, 118560, 2022.

## HONORS & AWARDS

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|---------------------------------------------------------------|------|
| Shanghai Super Postdoctoral Incentive Program                 | 2025 |
| China Postdoctoral Science Foundation (Grade C)               | 2025 |
| GPPS Excellent Paper Award (4/114)                            | 2024 |
| Outstanding Participant Award, Aerospace Doctoral Competition | 2024 |

## ACADEMIC SERVICE

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### Journal Reviewer:

- *Aerospace Science and Technology*
- *Physics of Fluids*
- *GPPS Journal*
- *Nuclear Engineering and Design*
- *International Journal of Turbo & Jet-Engines*
- *Journal of Engineering for Thermal Energy and Power*

### Academic Appointments:

- Session Chair, Compressor Session, GPPS Conference, 2023
- Early-Career Editorial Board Member, *Journal of Engineering for Thermal Energy and Power*